SMART INDIA HACKATHON 2026

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- **Problem Statement ID:** SIH26035
- **Problem Statement Title:** Development of a Software Program/Application for Generation of Test Reports for Non-Automatic Weighing Instruments (NAWI) as per OIML Recommendation R- 76
- **Theme:** Miscellaneous
- **PS Category:** Software
- **Team ID:** 137649
- **Team Name:** Kalpataru



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Idea Title

Voice-first AI commerce infrastructure for traditional artisans

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PROPOSED SOLUTION:

HOW IT ADDRESSES THE PROBLEM :

WORKFLOW DIAGRAM:

1 **User**



2 **Onboard**
for 1st time

Home Screen
Saved Login

3 **Product Image**
Guidlines based

Voice Input
AI Assisted

4 **Generated Catalogue**



Listing's Lifecycle
Track & Analyse

7 **Publish & Share**
Marketplaces & Social media

6 **Price + GTM E-commerce Posts**
Through Voice + visual cues

5 **Review & Edit**
Through Voice + visual cues



INNOVATION AND UNIQUENESS OF THE SOLUTION :

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TECHNICAL APPROACH

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TECH STACK:

ARCHITECTURE DIAGRAM :

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FEASIBILITY AND VIABILITY

WORKING PROTOTYPE • CONTROLLED AI • MODULAR SCALE

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TECHNICAL FEASIBILITY

ECONOMIC FEASIBILITY

USER VIABILITY

OPERATIONAL FEASIBILITY

SCALABILITY

FINANCIAL VIABILITY

Challenges	Strategy
Dialect & Noise Variations in Rural Speech	Real-time audio playback + visual transcript confirmation; ASR confidence threshold > 0.80 triggers manual/audio correction.
AI Hallucinations in Heritage Storytelling	Strict structural separation of artisanProvidedFacts and researchFacts; unverified claims are rejected.
Image Alteration / Loss of Handmade Texture	Automated artifact-fidelity check (>0.93 threshold); original raw photo is immutably archived.
Artisan Skepticism & Distrust of Pricing	Explainable price breakdown showing raw materials + labor wages + market position; artisan retains 100% final pricing control.

Processing Stage	Latency	Technology / Mathematical Model
Deterministic Blur & Quality Analysis	**4.2 ms**	Structure Tensor Sobel Eigenvalues
CIE LAB Softbox Studio Lighting	**38.6 ms**	Vectorized NumPy / CLAHE Kernel
Nonlinear Deblurring (Optional)	**312.0 ms**	Quantized CPU/ONNX NAFNet
Vernacular Speech Recognition (30s Audio)	**420.0 ms**	Groq LPU Whisper Large-v3
Multimodal Craft Fact Extraction	**850.0 ms**	Pydantic Schema-Guided VLM

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IMPACT AND BENEFITS

7M+ Artisans in India's handicrafts sector

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POTENTIAL IMPACT ON THE TARGET AUDIENCE

BENEFITS OF THE SOLUTION

Benefits	Value Delivered
Accessible Digital Commerce	Simple, minimalist UI designed for low digital literacy.
Professional Product Listings	Image + regional voice → AI-generated, e-commerce-ready catalogue.
Smart & Competitive Pricing	Price suggestions based on market trends + raw-material costs.
On-Demand Market Access	Turns buyer requirements into direct opportunities for artisans to fulfill.
Schemes & Events in One Place	Easy access to government schemes, exhibitions & trade fairs.
Wider Market Connectivity	One catalogue connects B2B buyers & government e-marketplaces.
Virtual Business Manager	Supports the artisan end-to-end, from product selling to managing orders and ongoing business needs.

SELF-SUSTAINING BUSINESS MODEL

FUTURE SCOPE

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RESEARCH AND REFERENCES

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RESEARCH VALIDATION

COMPETITIVE FEATURE GAP ANALYSIS

What the SIH PS demands	Kalpataru	EcomCatalog AI	India Handmade	Gaatha
Voice-First Regional Cataloging	✓	✓	×	×
AI Craft Image Studio (Fidelity > 0.93)	✓	×	×	×
Craft-aware dynamic price recommendation	✓	×	×	×
Automated GI Tagging & Geo-Provenance	✓	×	✓	×
Offline-Tolerant Mobile Workflow	✓	×	△	✓
Direct MoSJE / GeM / ODOP Scheme Bridge	✓	△	△	×
Product + artisan story + cultural identity	✓	×	△	✓

✓ = Available | △ = Partial/Limited | × = Not a core offering